

## EDSBY, Sweden

WMO: 023380

Lat: 61.3607N

Lon: 15.7144E

Elev: 185

StdP: 99.12

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -24.3      | -21.3 | -26.7                         | 0.3 | -24.1 | -23.6 | 0.5 | -21.0 | 9.7                   | 1.4  | 7.3 | 0.8  | 1.6                   | 220  | 0.622 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.1                   | 25.9            | 17.4 | 24.0 | 16.5 | 22.2 | 15.7 | 18.6                | 23.6 | 17.6 | 22.3 | 16.6 | 20.6 | 2.7                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.9                            | 12.4 | 19.9 | 15.8 | 11.5 | 19.3 | 14.8 | 10.7 | 18.2 | 53.4          | 23.5 | 50.3 | 22.4 | 47.1 | 20.7 | 22.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 7.0               | 5.8  | 5.1 | -28.2                      | 28.3 | 3.4                | 1.7 | -30.6                                              | 29.5 | -32.6      | 30.5 | -34.5      | 31.5 | -37.0      | 32.7 |
|                   |      |     | -28.1                      | 20.1 | 3.1                | 1.4 | -30.3                                              | 21.1 | -32.2      | 21.9 | -33.9      | 22.7 | -36.2      | 23.8 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 3.9    | -6.8  | -5.7  | -2.2  | 3.1   | 8.4   | 12.4  | 15.3  | 13.8  | 9.5   | 4.0   | -0.9  | -4.9  |  |  |
|                                                                        | DBStd       | 8.90   | 6.71  | 6.32  | 5.22  | 3.37  | 3.65  | 3.17  | 2.66  | 3.03  | 3.26  | 3.88  | 4.73  | 6.66  |  |  |
|                                                                        | HDD10.0     | 2663   | 520   | 439   | 379   | 208   | 75    | 12    | 1     | 4     | 48    | 189   | 326   | 462   |  |  |
|                                                                        | HDD18.3     | 5283   | 778   | 673   | 638   | 457   | 308   | 178   | 100   | 144   | 266   | 445   | 576   | 720   |  |  |
|                                                                        | CDD10.0     | 432    | 0     | 0     | 0     | 1     | 25    | 84    | 165   | 122   | 32    | 3     | 0     | 0     |  |  |
|                                                                        | CDD18.3     | 10     | 0     | 0     | 0     | 0     | 0     | 1     | 6     | 3     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 227    | 0     | 0     | 0     | 0     | 11    | 42    | 124   | 50    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 24     | 0     | 0     | 0     | 0     | 0     | 3     | 17    | 4     | 0     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.2    | 2.2   | 2.2   | 2.4   | 2.3   | 2.4   | 2.2   | 2.0   | 1.9   | 2.1   | 2.1   | 2.0   | 2.1   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 598    | 36    | 28    | 26    | 33    | 49    | 68    | 85    | 76    | 51    | 53    | 50    | 39    |  |  |
|                                                                        | PrecMax     | 947    | 67    | 54    | 51    | 84    | 98    | 142   | 243   | 143   | 127   | 139   | 145   | 75    |  |  |
|                                                                        | PrecMin     | 421    | 7     | 10    | 3     | 7     | 14    | 22    | 33    | 36    | 20    | 0     | 15    | 5     |  |  |
|                                                                        | PrecStd     | 112    | 17    | 12    | 12    | 18    | 22    | 29    | 48    | 27    | 28    | 36    | 31    | 17    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 5.8   | 7.2   | 12.0  | 18.6  | 25.0  | 27.0  | 28.5  | 27.1  | 21.4  | 14.8  | 10.8  | 8.3   |  |  |
|                                                                        |             | MCWB   | 3.9   | 4.0   | 6.4   | 10.9  | 15.0  | 16.0  | 18.8  | 18.1  | 15.2  | 11.7  | 9.1   | 6.6   |  |  |
|                                                                        | 2%          | DB     | 4.4   | 5.4   | 8.7   | 15.2  | 22.0  | 24.4  | 26.5  | 24.8  | 19.2  | 12.9  | 8.9   | 6.1   |  |  |
|                                                                        |             | MCWB   | 3.0   | 3.2   | 4.5   | 9.0   | 13.5  | 15.5  | 18.3  | 17.4  | 14.2  | 10.4  | 7.5   | 4.9   |  |  |
|                                                                        | 5%          | DB     | 3.1   | 4.0   | 6.7   | 12.6  | 19.3  | 22.1  | 24.5  | 22.5  | 17.2  | 11.5  | 7.0   | 4.6   |  |  |
|                                                                        |             | MCWB   | 1.9   | 2.2   | 3.2   | 7.3   | 12.2  | 14.7  | 17.3  | 16.4  | 13.3  | 9.6   | 5.9   | 3.3   |  |  |
|                                                                        | 10%         | DB     | 1.8   | 2.6   | 5.1   | 10.3  | 16.6  | 19.8  | 22.5  | 20.5  | 15.6  | 10.2  | 5.5   | 3.1   |  |  |
|                                                                        |             | MCWB   | 0.8   | 1.0   | 2.3   | 5.9   | 10.6  | 13.7  | 16.4  | 15.5  | 12.4  | 8.6   | 4.6   | 2.0   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 4.6   | 4.7   | 6.7   | 11.8  | 16.0  | 18.1  | 20.5  | 19.5  | 16.4  | 12.4  | 9.4   | 6.9   |  |  |
|                                                                        |             | MCDB   | 5.6   | 6.3   | 11.2  | 17.4  | 22.4  | 23.9  | 25.2  | 23.4  | 19.9  | 14.0  | 10.5  | 8.1   |  |  |
|                                                                        | 2%          | WB     | 3.1   | 3.5   | 5.0   | 9.4   | 14.4  | 16.5  | 19.2  | 18.2  | 15.0  | 11.1  | 7.6   | 5.0   |  |  |
|                                                                        |             | MCDB   | 4.2   | 5.1   | 8.0   | 14.5  | 20.7  | 21.8  | 24.2  | 22.8  | 18.0  | 12.3  | 8.7   | 5.9   |  |  |
|                                                                        | 5%          | WB     | 2.1   | 2.4   | 3.8   | 7.7   | 12.7  | 15.4  | 18.1  | 17.2  | 14.0  | 9.9   | 6.1   | 3.4   |  |  |
|                                                                        |             | MCDB   | 3.1   | 3.9   | 6.3   | 11.7  | 18.3  | 20.6  | 23.1  | 21.3  | 16.6  | 11.2  | 6.9   | 4.4   |  |  |
|                                                                        | 10%         | WB     | 0.8   | 1.2   | 2.5   | 6.3   | 11.2  | 14.4  | 17.1  | 16.2  | 12.9  | 8.7   | 4.7   | 2.1   |  |  |
|                                                                        |             | MCDB   | 1.6   | 2.5   | 4.8   | 9.6   | 16.1  | 19.1  | 21.4  | 19.7  | 14.9  | 10.1  | 5.4   | 3.0   |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 7.4   | 8.8   | 10.2  | 10.5  | 11.5  | 10.8  | 11.1  | 11.0  | 10.0  | 7.4   | 5.6   | 6.4   |  |  |
|                                                                        | 5% DB       | MCDBR  | 6.1   | 7.9   | 11.2  | 15.9  | 17.4  | 15.9  | 15.8  | 14.7  | 12.6  | 8.7   | 6.7   | 6.0   |  |  |
|                                                                        |             | MCWBR  | 5.5   | 6.3   | 7.5   | 9.8   | 9.6   | 8.0   | 8.1   | 8.2   | 8.1   | 6.6   | 6.0   | 5.4   |  |  |
|                                                                        | 5% WB       | MCDBR  | 5.9   | 7.4   | 10.4  | 14.2  | 15.3  | 13.4  | 13.6  | 12.7  | 10.8  | 7.6   | 6.3   | 5.7   |  |  |
|                                                                        |             | MCWBR  | 5.5   | 6.1   | 7.3   | 9.1   | 8.7   | 7.2   | 7.5   | 7.4   | 7.4   | 6.2   | 5.8   | 5.2   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.238 | 0.289 | 0.314 | 0.349 | 0.352 | 0.347 | 0.365 | 0.358 | 0.324 | 0.310 | 0.259 | 0.348 |  |  |
|                                                                        | taud        |        | 2.162 | 2.283 | 2.350 | 2.385 | 2.435 | 2.478 | 2.441 | 2.457 | 2.535 | 2.468 | 2.231 | 2.698 |  |  |
|                                                                        | Ebn at Noon |        | 498   | 694   | 808   | 845   | 871   | 881   | 852   | 829   | 800   | 675   | 464   | 343   |  |  |
|                                                                        | Edh at Noon |        | 35    | 62    | 82    | 97    | 100   | 98    | 99    | 90    | 70    | 53    | 33    | 23    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0.26  | 0.93  | 2.51  | 4.00  | 4.75  | 5.37  | 4.94  | 3.78  | 2.45  | 1.13  | 0.34  | 0.15  |  |  |
|                                                                        | RadStd      |        | 0.04  | 0.17  | 0.25  | 0.40  | 0.50  | 0.40  | 0.58  | 0.35  | 0.33  | 0.19  | 0.08  | 0.02  |  |  |

## Historical Trends

|                              |  | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (41 neighbors) |  | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page